*4.23 Repeat Problem 4.22 for $\mathbf{D} = \hat{\mathbf{x}} xy^3z^3$ (C/m^2).

4.24 Charge Q_1 is uniformly distributed over a thin spherical shell of radius a, and charge Q_2 is uniformly distributed over a second spherical shell of radius b with $b > a$. Apply Gauss's law to find $\mathbf{E}$ in the regions $R < a$, $a < R < b$, and $R > b$.

*4.25 The electric flux density inside a dielectric sphere of radius a centered at the origin is given by

$$\mathbf{D} = \hat{\mathbf{R}} \rho_0 R \qquad (\text{C/m}^2)$$

where ρ_0 is a constant. Find the total charge inside the sphere.

4.26 In a certain region of space, the charge density is given in cylindrical coordinates by the function

$$\rho_v = 5re^{-r} \quad (\text{C/m}^3).$$

Apply Gauss's law to find $\mathbf{D}$.

*4.27 An infinitely long cylindrical shell extending between $r = 1$ m and $r = 3$ m contains a uniform charge density ρ_{v0}. Apply Gauss's law to find $\mathbf{D}$ in all regions.

4.28 If the charge density increases linearly with distance from the origin such that $\rho_v = 0$ at the origin and $\rho_v = 4$ C/m^3 at $R = 2$ m, find the corresponding variation of $\mathbf{D}$.

4.29 A spherical shell with outer radius b surrounds a charge-free cavity of radius $a < b$ (Fig. P4.29). If the shell contains a charge density given by

$$\rho_v = -\frac{\rho_{v0}}{R^2}, \qquad a \le R \le b,$$

where ρ_{v0} is a positive constant, determine $\mathbf{D}$ in all regions.

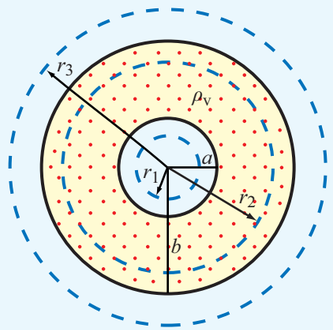


Figure P4.29 Problem 4.29.

Section 4-5: Electric Potential

*4.30 A square in the x–y plane in free space has a point charge of $+Q$ at corner $(a/2, a/2)$, the same at corner $(a/2, -a/2)$, and a point charge of $-Q$ at each of the other two corners.

(a) Find the electric potential at any point P along the x axis.

(b) Evaluate V at $x = a/2$.

4.31 The circular disk of radius a shown in Fig. 4-7 has uniform charge density ρ_s across its surface.

(a) Obtain an expression for the electric potential V at a point $P(0, 0, z)$ on the z axis.

(b) Use your result to find $\mathbf{E}$ and then evaluate it for $z = h$. Compare your final expression with (4.24), which was obtained on the basis of Coulomb's law.

*4.32 A circular ring of charge of radius a lies in the x–y plane and is centered at the origin. Assume also that the ring is in air and carries a uniform density ρ_ℓ.

(a) Show that the electrical potential at $(0, 0, z)$ is given by $V = \rho_\ell a / [2\varepsilon_0 (a^2 + z^2)^{1/2}]$.

(b) Find the corresponding electric field $\mathbf{E}$.

4.33 Show that the electric potential difference V_{12} between two points in air at radial distances r_1 and r_2 from an infinite line of charge with density ρ_ℓ along the z axis is $V_{12} = (\rho_\ell / 2\pi\varepsilon_0) \ln(r_2/r_1)$.

*4.34 Find the electric potential V at a location a distance b from the origin in the x–y plane due to a line charge with charge density ρ_ℓ and of length l. The line charge is coincident with the z axis and extends from $z = -l/2$ to $z = l/2$.

4.35 For the electric dipole shown in Fig. 4-15, $d = 1$ cm and $|\mathbf{E}| = 4$ (mV/m) at $R = 1$ m and $\theta = 0°$. Find $\mathbf{E}$ at $R = 2$ m and $\theta = 90°$.

*4.23 对 $\mathrm{D} = \hat{\mathrm{x}} \mathrm{xy}^3\mathrm{z}^3$ (C/m^2) 重复问题 4.22。

4.24 电荷 Q_1 均匀分布在半径为 a 的薄球壳上，电荷 Q_2 均匀分布在半径为 b 且 $b > a$ 的第二个球壳上。应用高斯定律求出区域 $R < a$、$a < R < b$ 和 $R > b$ 中的 E。

*4.25 以原点为中心、半径为 a 的电介质球内的电通量密度由下式给出

$$\mathbf{D} = \hat{\mathbf{R}} \rho_0 R \qquad (\text{C/m}^2)$$

其中 ρ_0 是常数。求球体内部的总电荷。

4.26 在空间的某个区域，电荷密度在柱坐标系中由函数给出

$$\rho_v = 5re^{-r} \quad (\text{C/m}^3).$$

应用高斯定律求 D。

*4.27 在 $r = 1$ m 和 $r = 3$ m 之间延伸的无限长圆柱壳包含均匀电荷密度 ρ_{v0}。应用高斯定律求出所有区域中的 D。

4.28 如果电荷密度随着距原点的距离线性增加，使得原点处的 $\rho_v = 0$ 和 $R = 2$ m 处的 $\rho_v = 4$ C/m^3，求 D 的相应变化。

4.29 外半径为 b 的球壳包围着半径为 $a < b$ 的无电荷空腔（图 P4.29）。如果外壳包含电荷密度由下式给出

$$\rho_v = -\frac{\rho_{v0}}{R^2}, \qquad a \le R \le b,$$

其中ρ_{v0}是正常数，确定所有区域中的D。

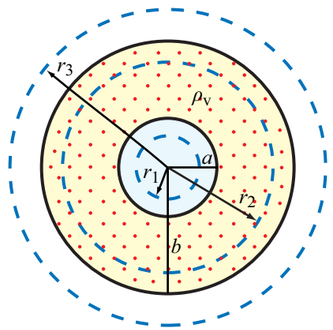


图 P4.29 问题 4.29。

第 4-5 节：电势

*4.30 自由空间中 x-y 平面上的正方形在角点 $(a/2, a/2)$ 处具有 +Q 的点电荷，在角点 $(a/2, -a/2)$ 处具有相同的点电荷，在其他两个角处各有一个点电荷 −Q。

(a) 求沿 x 轴任意点 P 的电势。 (b) 评估 $x = a/2$ 处的 V。

4.31 图 4-7 所示的半径为 a 的圆盘在其表面上具有均匀的电荷密度 ρ_s。

(a) 求出z轴上点P(0,0,z)处的电势V的表达式。 (b) 使用结果求出 E，然后计算它的 $z = h$。将你的最终表达式与根据库仑定律获得的 (4.24) 进行比较。

*4.32 半径为 a 的电荷圆环位于 x-y 平面内，且中心位于原点。还假设环在空气中并且具有均匀的密度ρ_ℓ。

(a) 证明 (0, 0, z) 处的电势由 $V = \rho_\ell a / [2\varepsilon_0 (a^2 + z^2)^{1/2}]$ 给出。

(b) 求相应的电场E。

4.33 证明沿 z 轴，距离密度为 ρ_ℓ 的无限电荷线，空气中距离 r_1 和 r_2 的两点之间的电势差 V_{12} 为 $V_{12} = (\rho_\ell / 2\pi\varepsilon_0) \ln(r_2/r_1)$。

*4.34 求 x-y 平面中距原点距离 b 的位置处由于电荷密度为 ρ_ℓ 且长度为 l 的线电荷而产生的电势 V。线电荷与 z 轴重合，并从 $z = -l/2$ 延伸到 $z = l/2$。

4.35 对于图 4-15 所示的电偶极子，在 $R = 1$ m 和 $\theta = 0°$ 处，$d = 1$ cm 和 $|E| = 4$ (mV/m)。在 $R = 2$ m 和 $\theta = 90°$ 处找到 E。

4.36 For each of the distributions of the electric potential V shown in **Fig. P4.36**, sketch the corresponding distribution of $\mathbf{E}$ (in all cases, the vertical axis is in volts and the horizontal axis is in meters).

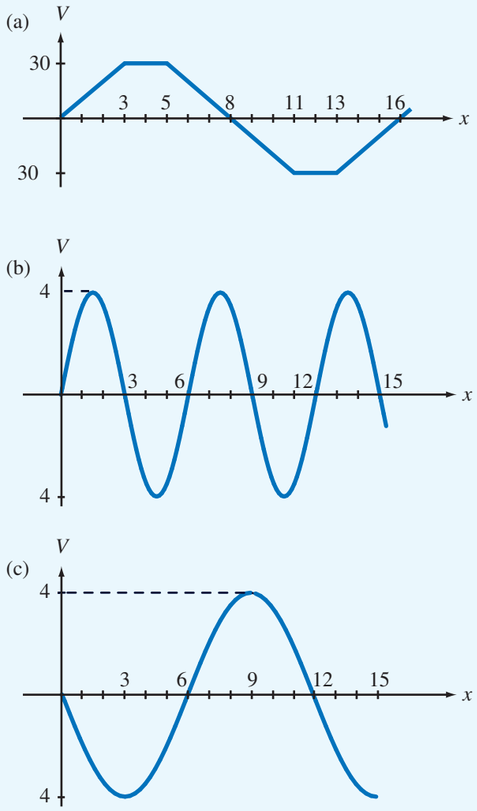


Figure P4.36 Electric potential distributions of Problem 4.36.

*4.37 As shown in **Fig. P4.37**, two infinite lines of charge, both parallel to the z axis, lie in the x–z plane: one with density ρ_ℓ and located at $x = a$ and the other with density $-\rho_\ell$ and located at $x = -a$. Obtain an expression for the electric potential $V(x, y)$ at a point $P = (x, y)$ relative to the potential at the origin.

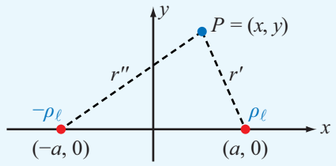


Figure P4.37 Problem 4.37.

4.38 Given the electric field

$$\mathbf{E} = \hat{\mathbf{R}}\,\frac{18}{R^2} \quad \text{(V/m)}$$

find the electric potential of point A with respect to point B where A is at $+2$ m and B at -4 m (both on the z axis).

*4.39 An infinitely long line of charge with uniform density $\rho_l = 9$ (nC/m) lies in the x–y plane parallel to the y axis at $x = 2$ m. Find the potential V_{AB} at point $A(3\text{ m}, 0, 4\text{ m})$ in Cartesian coordinates with respect to point $B(0,0,0)$ by applying the result of Problem 4.33.

4.40 The x–y plane contains a uniform sheet of charge with $\rho_{s_1} = 0.2$ (nC/m^2). A second sheet with $\rho_{s_2} = -0.2$ (nC/m^2) occupies the plane $z = 6$ m. Find V_{AB}, V_{BC}, and V_{AC} for $A(0,0,6\text{ m})$, $B(0,0,0)$, and $C(0,-2\text{ m},2\text{ m})$.

Section 4-6: Conductors

4.41 A cylindrical bar of silicon has a radius of 4 mm and a length of 8 cm. If a voltage of 5 V is applied between the ends of the bar and $\mu_e = 0.13$ (m^2/V·s), $\mu_h = 0.05$ (m^2/V·s), $N_e = 1.5 \times 10^{16}$ electrons/m^3, and $N_h = N_e$, find the following:

(a) the conductivity of silicon,

(b) the current I flowing in the bar,

*(c) the drift velocities $\mathbf{u}_e$ and $\mathbf{u}_h$,

(d) the resistance of the bar, and

(e) the power dissipated in the bar.

4.42 Repeat Problem 4.41 for a bar of germanium with $\mu_e = 0.4$ (m^2/V·s), $\mu_h = 0.2$ (m^2/V·s), and $N_e = N_h = 2.4 \times 10^{19}$ electrons or holes/m^3.

4.43 A 100-m long conductor of uniform cross-section has a voltage drop of 4 V between its ends. If the density of the current flowing through it is 1.4×10^6 (A/m^2), identify the material of the conductor.

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4.36 对于图 P4.36 所示的每种电势 V 的分布，画出相应的 E 分布（在所有情况下，纵轴的单位是伏特，横轴的单位是米）。

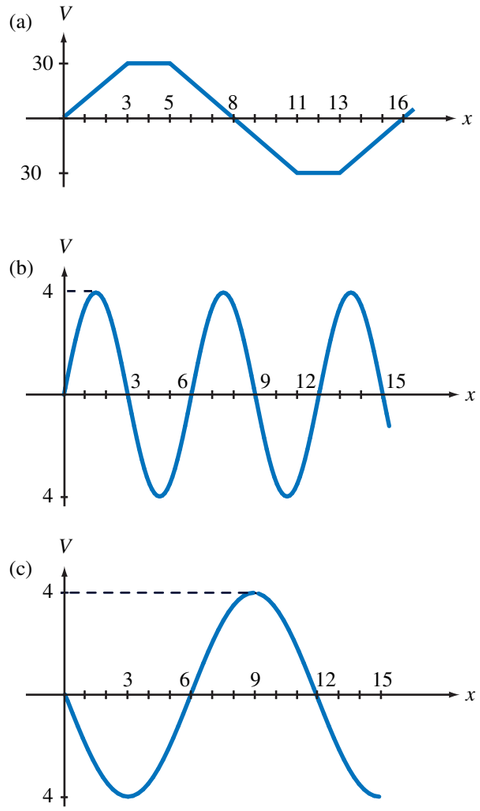


图 P4.36 问题 4.36 的电势分布。

*4.37 如图 P4.37 所示，两条无限电荷线均平行于 z 轴，位于 x-z 平面内：一条密度为 ρ_ℓ，位于 x = a，另一条密度为 $-\rho_\ell$，位于 x = −a。求出 P= (x,y) 点处的电势 V(x,y) 相对于原点电势的表达式。

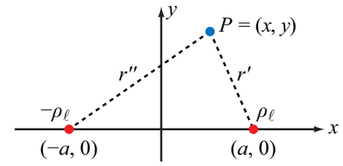


图 P4.37 问题 4.37。

4.38 给定电场

$$\mathbf{E} = \hat{\mathbf{R}}\,\frac{18}{R^2} \quad \text{(V/m)}$$

求 A 点相对于 B 点的电势，其中 A 位于 +2 m，B 位于 −4 m（均在 z 轴上）。

*4.39 一条具有均匀密度 $\rho_l = 9$ (nC/m) 的无限长电荷线位于 x-y 平面内，平行于 y 轴，位于 x = 2 m 处。应用问题 4.33 的结果，求直角坐标系中 A(3 m, 0, 4 m) 点相对于 B(0, 0, 0) 点的势 V_{AB}。

4.40 x–y 平面包含 $\rho_{s_1} = 0.2$ (nC/m^2) 的均匀电荷层。$\rho_{s_2} = -0.2$ (nC/m^2) 的第二片占据平面 z = 6 m。求 A(0, 0, 6 m)、B(0, 0, 0) 和 C(0, −2 m, 2 m) 的 V_{AB}、V_{BC} 和 V_{AC}。

第 4-6 节：导体

4.41 圆柱形硅棒的半径为 4 毫米，长度为 8 厘米。如果在棒两端之间施加 5 V 电压，$\mu_e = 0.13$ (m^2/V·s)、$\mu_h = 0.05$ (m^2/V·s)、$N_e = 1.5 \times 10^{16}$ 电子/m^3 和 $N_h = N_e$，找到以下内容：

(a) 硅的电导率，(b) 流过硅棒的电流 I，

*(c) 漂移速度 u_e 和 u_h，(d) 棒的电阻，以及 (e) 棒中消耗的功率。

4.42 对具有 $\mu_e = 0.4$ (m^2/V·s)、$\mu_h = 0.2$ (m^2/V·s) 和 $N_e = N_h = 2.4 \times 10^{19}$ 个电子或空穴/m^3 的锗棒重复问题 4.41。

4.43 一根 100 米长的均匀横截面导体，其两端之间的电压降为 4 V。如果流过它的电流密度为 1.4×10^6 (A/m^2)，则确定导体的材料。

4.44 A coaxial resistor of length l consists of two concentric cylinders. The inner cylinder has radius a and is made of a material with conductivity σ_1, and the outer cylinder, extending between $r = a$ and $r = b$, is made of a material with conductivity σ_2. If the two ends of the resistor are capped with conducting plates, show that the resistance between the two ends is $R = l/[\pi(\sigma_1 a^2 + \sigma_2(b^2 - a^2))]$.

*__4.45__ Apply the result of Problem 4.44 to find the resistance of a 20 cm long hollow cylinder (Fig. P4.45) made of carbon with $\sigma = 3 \times 10^4$ (S/m).

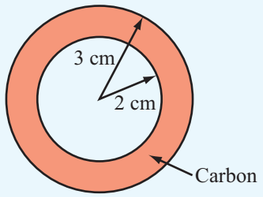


Figure P4.45 Cross-section of hollow cylinder of Problem 4.45.

4.46 A 2×10^{-3} mm thick square sheet of aluminum has 5 cm $\times$ 5 cm faces. Find the following:

(a) The resistance between opposite edges on a square face.

(b) The resistance between the two square faces. (See Appendix B for the electrical constants of materials.)

4.47 A cylinder-shaped carbon resistor is 8 cm in length, and its circular cross section has a diameter $d = 1$ mm.

(a) Determine the resistance R.

(b) To reduce its resistance by 40%, the carbon resistor is coated with a layer of copper of thickness t. Use the result of Problem 4.44 to determine t.

Section 4-8: Boundary Conditions

*__4.48__ With reference to Fig. 4-21, find $\mathbf{E}_1$ if

$$\mathbf{E}_2 = \hat{\mathbf{x}}3 - \hat{\mathbf{y}}2 + \hat{\mathbf{z}}2 \quad \text{(V/m)},$$

$\varepsilon_1 = 2\varepsilon_0$, $\varepsilon_2 = 18\varepsilon_0$, and the boundary has a surface charge density $\rho_s = 3.54 \times 10^{-11}$ (C/m^2). What angle does $\mathbf{E}_2$ make with the z axis?

4.49 An infinitely long cylinder of radius a is surrounded by a dielectric medium that contains no free charges. If the tangential component of the electric field in the region $r \geq a$ is given by $\mathbf{E}_t = -\hat{\boldsymbol{\phi}} \cos\phi / r^2$, find $\mathbf{E}$ in that region.

*__4.50__ If $\mathbf{E} = \hat{\mathbf{R}}150$ (V/m) at the surface of a 5 cm conducting sphere centered at the origin, what is the total charge Q on the sphere's surface?

4.51 **Figure P4.51** shows three planar dielectric slabs of equal thickness but with different dielectric constants. If $\mathbf{E}_0$ in air makes an angle of 45° with respect to the z axis, find the angle of $\mathbf{E}$ in each of the other layers.

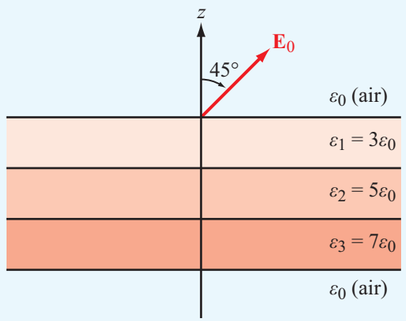


Figure P4.51 Dielectric slabs in Problem 4.51.

Sections 4-9 and 4-10: Capacitance and Electrical Energy

4.52 Determine the force of attraction in a parallel-plate capacitor with $A = 5$ cm^2, $d = 2$ cm, and $\varepsilon_r = 4$ if the voltage across it is 50 V.

4.53 Dielectric breakdown occurs in a material whenever the magnitude of the field $\mathbf{E}$ exceeds the dielectric strength anywhere in that material. In the coaxial capacitor of Example 4-15,

*__(a)__ At what value of r is $|E|$ maximum?

(b) What is the breakdown voltage if $a = 1$ cm, $b = 2$ cm, and the dielectric material is mica with $\varepsilon_r = 6$?

4.44 长度为 l 的同轴电阻器由两个同心圆柱体组成。内圆柱体具有半径a并且由具有电导率σ_1的材料制成，并且在r =a和r=b之间延伸的外圆柱体由具有电导率σ_2的材料制成。如果电阻器的两端都盖有导电片，则表明两端之间的电阻为$R = l/[\pi(\sigma_1 a^2 + \sigma_2(b^2 - a^2))]$。

*4.45 应用问题 4.44 的结果求出 20 厘米长的碳制空心圆柱体（图 P4.45）的电阻，其$\sigma = 3 \times 10^4$ (S/m)。

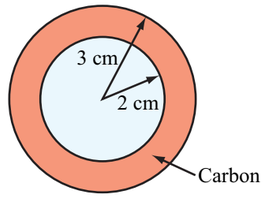


图 P4.45 问题 4.45 的空心圆柱体的横截面。

4.46 一块 2×10^{-3} 毫米厚的方形铝板具有 5 厘米 $\times$ 5 厘米的面。找到以下内容：

(a) 方形面上相对边缘之间的阻力。 (b) 两个正方形面之间的电阻。 （有关材料的电常数，请参阅附录 B。）

4.47 圆柱形碳电阻器长8厘米，其圆形横截面直径d=1毫米。

(a) 确定电阻R。

(b) 为了降低 40% 的电阻，碳电阻器上涂有一层厚度为 t 的铜。使用问题 4.44 的结果来确定 t。

第 4-8 节：边界条件

*__4.48__ 参考图 4-21，求 E_1 如果

$$\mathbf{E}_2 = \hat{\mathbf{x}}3 - \hat{\mathbf{y}}2 + \hat{\mathbf{z}}2 \quad \text{(V/m)},$$

$\varepsilon_1 = 2\varepsilon_0$、$\varepsilon_2 = 18\varepsilon_0$，边界表面电荷密度为$\rho_s = 3.54 \times 10^{-11}$ (C/m^2)。 E_2 与 z 轴成什么角度？

4.49 半径为 a 的无限长圆柱体被不含自由电荷的介电介质包围。如果 $r \geq a$ 区域中电场的切向分量由 $E_t = -\hat{\boldsymbol{\phi}} \cos\phi / r^2$ 给出，求该区域中的 E。

*4.50 如果 $E = \hat{R}150$ (V/m) 位于以原点为中心的 5 cm 导电球体表面，则该球体表面的总电荷 Q 是多少？

4.51 图 P4.51 显示了三个厚度相等但介电常数不同的平面电介质板。如果空气中的 E_0 与 z 轴成 45° 角，则求其他各层中 E 的角度。

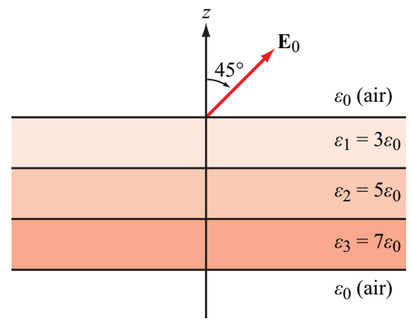


图 P4.51 问题 4.51 中的电介质板。

Sections 4-9 and 4-10: Capacitance and Electrical Energy

4.52 确定 $A = 5$ cm^2、$d = 2$ cm 和 $\varepsilon_r = 4$ 的平行板电容器（如果其两端电压为 50 V）的吸引力。

4.53 当电场 E 的大小超过材料中任何地方的介电强度时，材料中就会发生介电击穿。在实施例4-15的同轴电容器中，

*__(a)__ r 的值是多少时 |E| 最大？

(b) 如果a = 1 cm，b = 2 cm，介电材料为$\varepsilon_r = 6$ 云母，击穿电压是多少？

4.54 An electron with charge $Q_e = -1.6 \times 10^{-19}$ C and mass $m_e = 9.1 \times 10^{-31}$ kg is injected at a point adjacent to the negatively charged plate in the region between the plates of an air-filled, parallel-plate capacitor with separation of 1 cm and rectangular plates each 10 cm^2 in area (**Fig. P4.54**). If the voltage across the capacitor is 10 V, find the following:

(a) the force acting on the electron,

(b) the acceleration of the electron, and

(c) the time it takes the electron to reach the positively charged plate, assuming that it starts from rest.

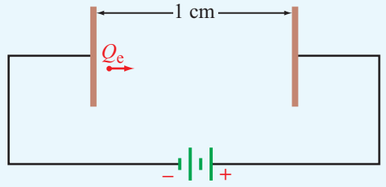
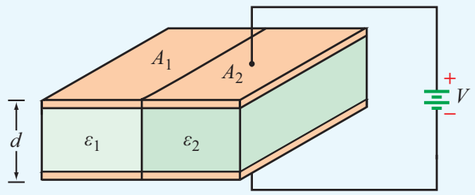
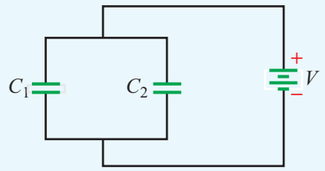


Figure P4.54 Electron between charged plates of Problem 4.54.



(a)



(b)

Figure P4.56 (a) Capacitor with parallel dielectric section and (b) equivalent circuit.

*__4.55__ In a dielectric medium with $\varepsilon_r = 4$, the electric field is given by

$$\mathbf{E} = \hat{\mathbf{x}}(x^2 + 2z) + \hat{\mathbf{y}}x^2 - \hat{\mathbf{z}}(y + z) \qquad \text{(V/m)}$$

Calculate the electrostatic energy stored in the region $-1 \text{ m} \le x \le 1 \text{ m}$, $0 \le y \le 2$ m, and $0 \le z \le 3$ m.

4.56 **Figure P4.56(a)** depicts a capacitor consisting of two parallel, conducting plates separated by a distance d. The space between the plates contains two adjacent dielectrics: one with permittivity ε_1 and surface area A_1 and another with ε_2 and A_2. The objective of this problem is to show that the capacitance C of the configuration shown in **Fig. P4.56(a)** is equivalent to two capacitances in parallel, as illustrated in **Fig. P4.56(b)**, with

$$C = C_1 + C_2, \qquad (4.146)$$

where

$$C_1 = \frac{\varepsilon_1 A_1}{d}, \qquad (4.147)$$

$$C_2 = \frac{\varepsilon_2 A_2}{d}. \qquad (4.148)$$

To this end, proceed as follows:

(a) Find the electric fields $\mathbf{E}_1$ and $\mathbf{E}_2$ in the two dielectric layers.

(b) Calculate the energy stored in each section and use the result to calculate C_1 and C_2.

(c) Use the total energy stored in the capacitor to obtain an expression for C. Show that Eq. (4.146) is indeed a valid result.

4.57 Use the result of Problem 4.56 to determine the capacitance for each of the following configurations.

*__(a)__ Conducting plates are on top and bottom faces of the rectangular structure in **Fig. P4.57(a)**.

(b) Conducting plates are on front and back faces of the structure in **Fig. P4.57(a)**.

(c) Conducting plates are on top and bottom faces of the cylindrical structure in **Fig. P4.57(b)**.

4.54 电荷为 $Q_e = -1.6 \times 10^{-19}$ C 且质量为 $m_e = 9.1 \times 10^{-31}$ kg 的电子被注入到空气填充的平行板电容器的板之间区域中与带负电的板相邻的点，该电容器的板间距为 1 cm，矩形板的面积各为 10 cm^2（图 P4.54）。如果电容器两端的电压为 10 V，则得出：

(a) 作用在电子上的力，(b) 电子的加速度，以及 (c) 电子到达带正电的板所需的时间（假设它从静止开始）。

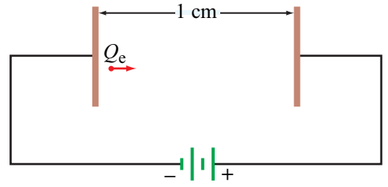
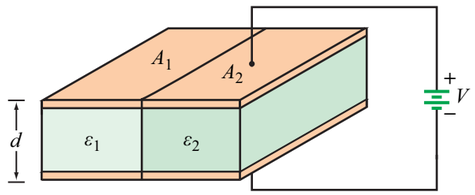
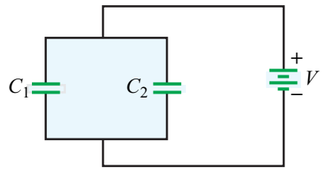


图 P4.54 问题 4.54 的带电板之间的电子。



(a)



(b)

图 P4.56 (a) 具有平行介电部分的电容器和 (b) 等效电路。

*4.55 在 $\varepsilon_r = 4$ 的介电介质中，电场由下式给出

$$\mathbf{E} = \hat{\mathbf{x}}(x^2 + 2z) + \hat{\mathbf{y}}x^2 - \hat{\mathbf{z}}(y + z) \qquad \text{(伏/米)}$$

计算区域 $-1 \text{ m} \le x \le 1$ m、$0 \le y \le 2$ m 和 $0 \le z \le 3$ m 中存储的静电能。

(a) 求两个介电层中的电场 E_1 和 E_2。 (b) 计算每个部分存储的能量，并使用结果计算 C_1 和 C_2。 (c) 使用电容器中存储的总能量来获得 C 的表达式。 (4.146) 确实是一个有效的结果。

4.56 图 P4.56(a) 描绘了一个由两个平行的导电板组成的电容器，两个导电板之间的距离为 d。板之间的空间包含两个相邻的电介质：一个具有介电常数 ε_1 和表面积 A_1，另一个具有 ε_2 和 A_2。该问题的目的是证明图 P4.56(a) 所示配置的电容 C 相当于两个并联电容，如图 P4.56(b) 所示，其中

$$C = C_1 + C_2, \qquad (4.146)$$

在哪里

$$C_1 = \frac{\varepsilon_1 A_1}{d}, \qquad (4.147)$$

$$C_2 = \frac{\varepsilon_2 A_2}{d}. \qquad (4.148)$$

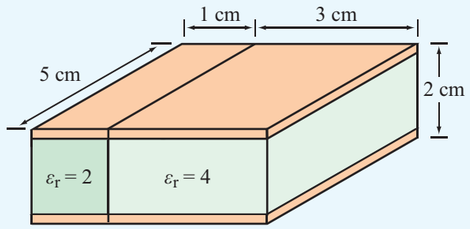
为此，请执行以下操作：

4.57 使用问题 4.56 的结果来确定以下每个配置的电容。

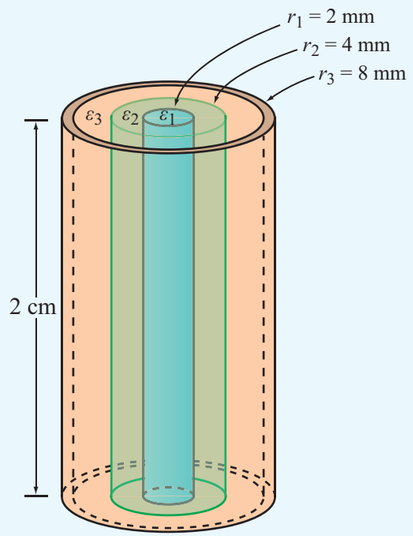
*(a) 导电板位于图P4.57(a)中矩形结构的顶面和底面。

(b) 导电板位于图 P4.57(a) 结构的正面和背面。

(c) 导电板位于图 P4.57(b) 中圆柱形结构的顶面和底面。



(a)



$\varepsilon_1 = 8\varepsilon_0$; $\varepsilon_2 = 4\varepsilon_0$; $\varepsilon_3 = 2\varepsilon_0$

(b)

Figure P4.57 Dielectric sections for Problems 4.57 and 4.59.

4.58 The capacitor shown in **Fig. P4.58** consists of two parallel dielectric layers. Use energy considerations to show that the equivalent capacitance of the overall capacitor, C, is equal to the series combination of the capacitances of the individual layers, C_1 and C_2, namely

$$C = \frac{C_1 C_2}{C_1 + C_2}, \tag{4.149}$$

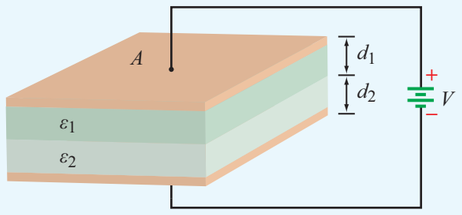
where

$$C_1 = \varepsilon_1 \frac{A}{d_1}, \qquad C_2 = \varepsilon_2 \frac{A}{d_2}.$$

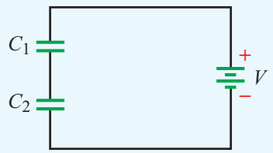
(a) Let V_1 and V_2 be the electric potentials across the upper and lower dielectrics, respectively. What are the corresponding electric fields E_1 and E_2? By applying the appropriate boundary condition at the interface between the two dielectrics, obtain explicit expressions for E_1 and E_2 in terms of ε_1, ε_2, and V and the indicated dimensions of the capacitor.

(b) Calculate the energy stored in each of the dielectric layers and then use the sum to obtain an expression for C.

(c) Show that C is given by Eq. (4.149).



(a)

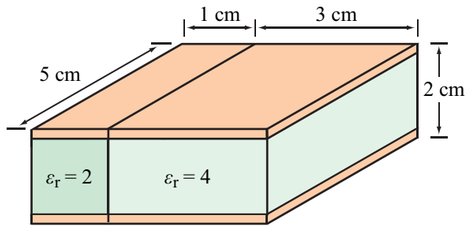


(b)

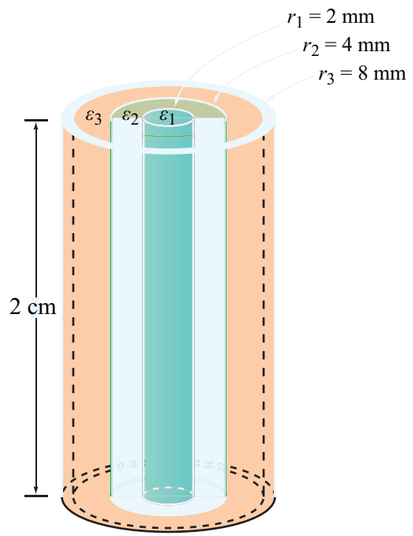
Figure P4.58 (a) Capacitor with parallel dielectric layers and (b) equivalent circuit (Problem 4.58).

4.59 Use the expressions given in Problem 4.58 to determine the capacitance for the configurations in **Fig. P4.57(a)** when the conducting plates are placed on the right and left faces of the structure.

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(a)



$\varepsilon_1 = 8\varepsilon_0$; $\varepsilon_2 = 4\varepsilon_0$; $\varepsilon_3 = 2\varepsilon_0$

(b)

图 P4.57 问题 4.57 和 4.59 的电介质部分。

4.58 图 P4.58 所示的电容器由两个平行的介电层组成。使用能量考虑表明整个电容器的等效电容 C 等于各层电容 C_1 和 C_2 的串联组合，即

$$C = \frac{C_1 C_2}{C_1 + C_2}, \tag{4.149}$$

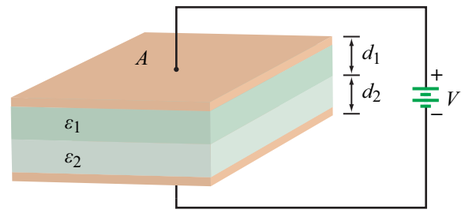
在哪里

$$C_1 = \varepsilon_1 \frac{A}{d_1}, \qquad C_2 = \varepsilon_2 \frac{A}{d_2}.$$

(a) 令 V_1 和 V_2 分别为上电介质和下电介质两端的电势。相应的电场 E_1 和 E_2 是什么？通过在两个电介质之间的界面处应用适当的边界条件，根据 ε_1、ε_2 和 V 以及电容器的指示尺寸获得 E_1 和 E_2 的显式表达式。

(b) 计算每个介电层中存储的能量，然后使用总和获得 C 的表达式。

(c) 证明 C 由方程式给出。（4.149）。



(a)



(b)

图 P4.58 (a) 具有平行介电层的电容器和 (b) 等效电路（问题 4.58）。

4.59 当导电板放置在结构的左右面上时，使用问题 4.58 中给出的表达式来确定图 P4.57(a) 中的配置的电容。